

7PAM2002 Data Science Project Module

Choosing a Project Form

Semester A 2025/2026

This form will be used by your supervisor to agree your project topic and dataset. Complete as much of the form as possible then submit the form into 'Assignments' on the Project Module Canvas site.

Student Full Name: shahab zaman

Student ID: 24056595

Email Address: shahabzaman718@gmail.com

Course: 1 year

Research (delete as appropriate)

Semester intake to the course: A

(delete as appropriate)

SECTION A	
What is your proposed project title or topic?	Physics Signal Classification: Separating Gamma Rays from Hadronic Backgrounds in Telescope Data
What is the Research Question for your project?	1. How robust are distance-based classifiers compared to boundary-based classifiers when noise is introduced to the sensor readings? 2. Can a simple linear model capture the decision boundary effectively, or is a non-linear approach strictly necessary? 3. How do Explainable AI methods interpret the contribution of different "Hillas parameters" (shape measurements) in distinguishing signal from background noise?
Dataset website address (or organisation and person the dataset is from).	UCI MAGIC Gamma Telescope
Where was the data originally collected? (who, when ,where)	The data was generated by a Monte Carlo program, Corsika, to simulate high-energy gamma particles in the MAGIC (Major Atmospheric Gamma Imaging Cherenkov) Telescope in La Palma, Canary Islands.
What type of data are you using? (e.g. image/tabular/category/continuous etc)	The dataset is Tabular. It consists of 10 Continuous/Numerical features representing "Hillas parameters" (geometric properties of the pulses/images) and a binary class label (Gamma vs. Hadron).
SECTION B	

Can you attend all 6 supervision sessions on-campus? If not state reason.	Yes
What was the subject of your BSc degree and any other Master's degrees you have taken?	BS in physics
What are your career aims and/or the industry sector you would like to get into?	My career aim is to build a meaningful career where I can continuously develop my skills, contribute to real-world impact, and grow into roles with greater responsibility.
A brief account of your programming/data science experience (including work or placement).	I have experienced in programming and data science through academic coursework and practical projects.
Any work experience (including non-computing related).	I do not yet have formal work experience. I am impatient to gain practical experience and apply my skills in a professional environment
Do you have any hobbies, activities or interests.	My hobbies and interests include exploring new technologies and improving my programming skills through personal projects and online learning.